

MECHANICAL ENGINEERING MATERIALS

318/2(N)

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- i) ☒ Coordination number of FCC structure:
 a) 6 b) 8 c) 10 ☒ 12
- ii) ☒ Pure iron at room temperature has:
 a) BCC b) FCC c) HCP d) SC
- iii) ☒ Atomic packing factor (APF) of BCC:
 a) 0.74 ☒ 0.68 c) 0.52 d) None of these
- iv) ☒ Which property enables a material to resist scratching, cutting and abrasion:
 ☒ Hardness b) Plasticity c) Fatigue d) Elasticity
- v) ☒ Toughness of a material when heated:
 a) Remains same ☒ Increases c) Decreases d) none of these
- vi) ☒ Hardest microstructure of steel:
 a) Ferrite ☒ Martensite c) Bainite d) Pearlite
- vii) ☒ Hardness of steel increases when % C:
 a) Decreases ☒ Increases c) All of the above d) None of the above
- viii) ☒ An eutectoid steel consists of:
 ☒ Wholly pearlite b) Wholly austenite c) Pearlite & ferrite d) Pearlite & cementite
- ix) ☒ In the Austempering process, austenite changes into:
 a) Martensite b) Troostite c) Sorbite ☒ Bainite
- x) ☒ In annealing, cooling is done in:
 a) Air ☒ Furnace c) Water d) None of these
- xi) ☒ Tempering is done after:
 ☒ Hardening b) Normalising c) Annealing d) None of these
- xii) ☒ Lower critical point for steels (°C):
 a) 600 ☒ 723 c) 700 d) 913
- xiii) ☒ Babbitt metal is used in:
 ☒ Bearings b) Gears c) Shafts d) None of these
- xiv) ☒ Long plastic pipes are produced by:
 a) Compression moulding ☒ Extrusion moulding c) Injection moulding d) Blow moulding
- xv) ☒ NDT is used to determine:
 ☒ Location of defects b) Chemical composition c) Corrosion of metal d) All of these

GROUP-B

Answer any Five (05) questions.

- 2.(a) What do you mean by a unit cell? (2)
- (b) For a Body-Centered Cubic (BCC) crystal structure, determine: (2+4)
 - (i) Average number of atoms per unit cell
 - (ii) Atomic packing factor

3. Define the following mechanical properties: i) Elasticity ii) Ductility iii) Malleability iv) Fatigue (2+2+2+2)
4. Draw the Iron-Carbon phase diagram and label all important points. (8)
5. (a) Write a short note on Cementite and Pearlite. (3 + 3)
 (b) Write the composition of High-Speed Steel (HSS). (2)
6. (a) Draw the TTT diagram for eutectoid steel showing all salient points. (5)
 (b) Write short notes on Case hardening (3)
7. (a) Describe the Annealing process. (4)
 (b) Write the purpose of Austempering and Martempering processes. (2 + 2)
8. (a) Write down the names of two copper-based alloys. (2)
 (b) Write the composition of Muntz Metal and Gun Metal. (2 + 2)
 (c) What are the applications of Monel Metal? (2)
9. (a) Compare Thermoplastics and Thermosetting plastics. (3)
 (b) With the help of a diagram, describe the Injection Moulding process. (5)
10. (a) What are the differences between Destructive and Non-Destructive Testing? (3)
 (b) Explain the Dye Penetration Test with a suitable sketch. (5)

